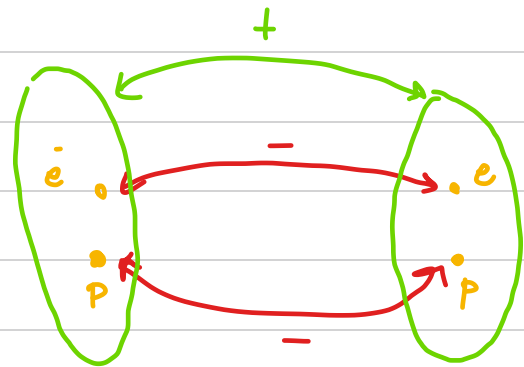


Composite Bosons & Fermions:

A single atom: $\psi(\vec{r}, \vec{R}) = u(m\vec{r} + M\vec{R}) v(\vec{R} - \vec{r})$



$$H = -\frac{\hbar^2}{2M} \nabla_{\vec{R}}^2 - \frac{\hbar^2}{2m} \nabla_{\vec{r}}^2 + V(|\vec{r} - \vec{R}|)$$

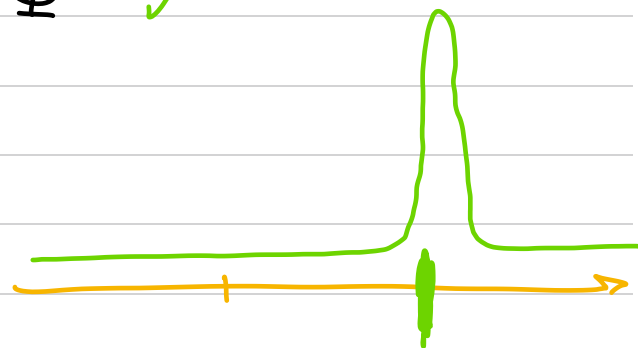
$$\vec{\Delta r} = \vec{R} - \vec{r}, \quad \vec{R}_{cm} := m\vec{r} + M\vec{R} \quad \rightarrow \quad H = \underbrace{H(\vec{R}_{cm})}_u + \underbrace{H(\vec{\Delta r})}_v \quad \rightarrow \quad \psi = u(\vec{R}_{cm}) v(\vec{\Delta r})$$

$$\bar{\Psi} = \psi_{11} \psi_{22} - \psi_{12} \psi_{21}$$

$$\psi_{jk} = u(M\vec{R}_j + m\vec{r}_k) v(\vec{R}_j - \vec{r}_k)$$

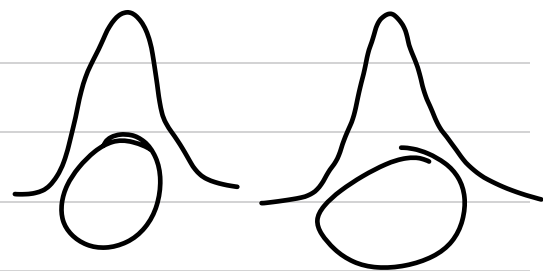
$$P_1 \rightarrow P_2 \quad \Psi \rightarrow \psi_{21} \psi_{12} - \psi_{22} \psi_{11} = -\Psi \quad \checkmark$$

$$e_1 \rightarrow e_2 \quad \Psi \rightarrow -\Psi \quad \checkmark$$



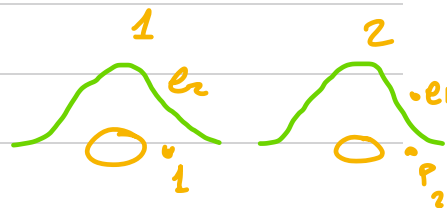
$$\psi(\vec{r}, \vec{R}) = \underbrace{u(m\vec{r} + M\vec{R})}_u \underbrace{v(\vec{R} - \vec{r})}_v$$

$v(\vec{r} - \vec{R})$ is highly localized around $\vec{r} = \vec{R}$



$$\bar{\Psi} = \psi_{11} \psi_{22} - \psi_{12} \psi_{21}$$

$$\psi_{jk} = u(M\vec{R}_j + m\vec{r}_k) v(\vec{R}_j - \vec{r}_k) \rightarrow \left\{ \begin{array}{l} \psi_{11} \psi_{22} : \vec{r}_1 \sim \vec{R}_1, \vec{r}_2 \sim \vec{R}_2 \\ \psi_{12} \psi_{21} : \vec{R}_1 \sim \vec{r}_2, \vec{R}_2 \sim \vec{r}_1 \end{array} \right\} \text{overlap } \vec{r}_1, \vec{r}_2 \sim \vec{R}_1 \sim \vec{R}_2$$



if Boson: $\psi_1 \psi_2 = u(M\vec{R}_1 + m\vec{r}_1) v(\vec{R}_1 - \vec{F}_1) u(M\vec{R}_2 + m\vec{r}_2) v(\vec{R}_2 - \vec{F}_2)$

two atoms in same state u, v

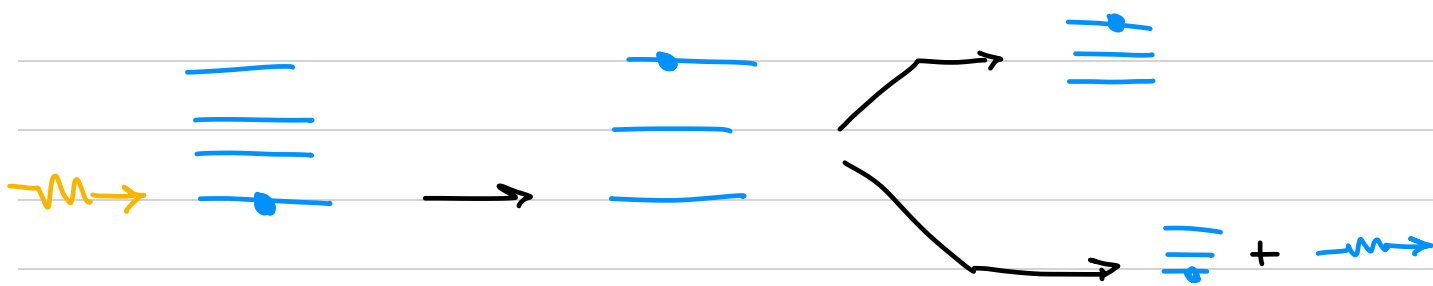
$$\Phi = \psi_{11} \psi_{22} - \psi_{12} \psi_{21}$$

$$\begin{aligned} \langle A \rangle &= \langle \Phi | \hat{A} | \Phi \rangle = \int d^3\vec{r}_1 d^3\vec{r}_2 d^3\vec{R}_1 d^3\vec{R}_2 (\psi_{11}^* \psi_{22}^* - \psi_{12}^* \psi_{21}^*) \hat{A} (\psi_{11} \psi_{22} - \psi_{12} \psi_{21}) \\ &= \int \dots A_{1122, 1122} + \underbrace{A_{1221, 1221}}_{A_{1122, 1122}} - \underbrace{A_{1221, 1122}}_{A_{1122, 1221}} - A_{1122, 1221} \\ &= 2 \int \dots (\underbrace{\psi_{11}^* \psi_{22}^* A \psi_{11} \psi_{22}}_{\text{Bosonic}} - \underbrace{\psi_{11}^* \psi_{22}^* A \psi_{12} \psi_{21}}_{\text{"remote"} \rightarrow 0}) \end{aligned}$$

$\Rightarrow$ atoms behave like bosons!

$$A_{1122, 1221} \rightarrow 0$$

$$\begin{array}{cc} \cdot e_1 & \cdot e_1 \\ \cdot & \cdot \\ \cdot p_1 & \cdot p_2 \end{array} \quad \begin{array}{cc} \cdot e_1 & \cdot e_2 \\ \cdot & \cdot \\ \cdot p_1 & \cdot p_2 \end{array}$$



$$|state\rangle = C_1 |excited\rangle + C_2 |ground + Photon\rangle$$

$$|\psi\rangle = \frac{|1\rangle \otimes |2\rangle + |2\rangle \otimes |1\rangle}{\sqrt{2}} \rightarrow \underbrace{|1\rangle}_{\text{state 1}} \underbrace{|1\rangle}_{\text{state 2}}$$

$$|\psi\rangle = \frac{|11\rangle \otimes |3\rangle - |3\rangle \otimes |1\rangle}{\sqrt{2}} \rightarrow |11010\rangle$$

4-level system

$|2031\rangle \rightarrow$ Bosonic

$$|n_1, n_2, n_3, \dots\rangle := \frac{1}{\sqrt{N}} \left(\underbrace{|1\rangle \otimes |1\rangle \otimes \dots \otimes |1\rangle}_{n_1} \otimes \underbrace{|2\rangle \otimes |2\rangle \otimes \dots \otimes |2\rangle}_{n_2} \otimes \dots + \text{Permutations} \right)$$

Fermion/Boson
↑

$$\frac{|110\rangle + |111\rangle}{\sqrt{2}}$$